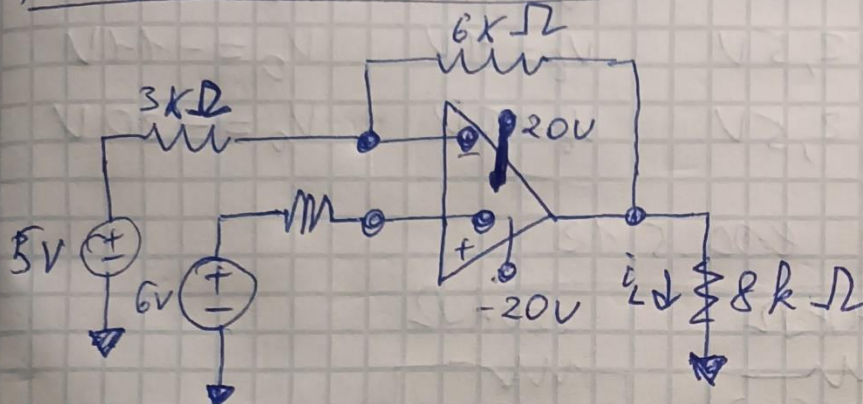


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HW #4

page 165 ex 153



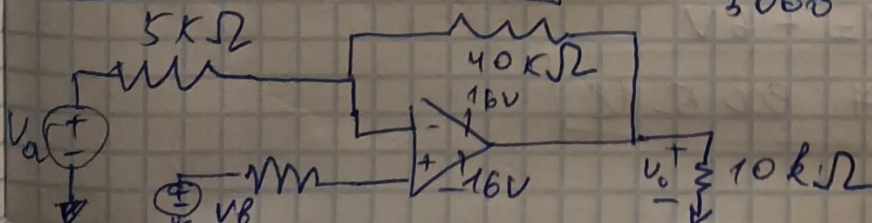
$$V_p = \frac{5000}{5000 + 10000} \cdot 6 = 2 \quad V = V_n$$

$$\frac{V_n + 5}{3000} + \frac{V_n - V_o}{8000} = 0$$

$$2(2 + 5) + (2 - V_o) = 0 \quad V_o = 16V$$

$$I_L = \frac{V_o}{8000} = \frac{16}{8000} = 2000 \cdot 10^{-6} \quad I_L = 2mA$$

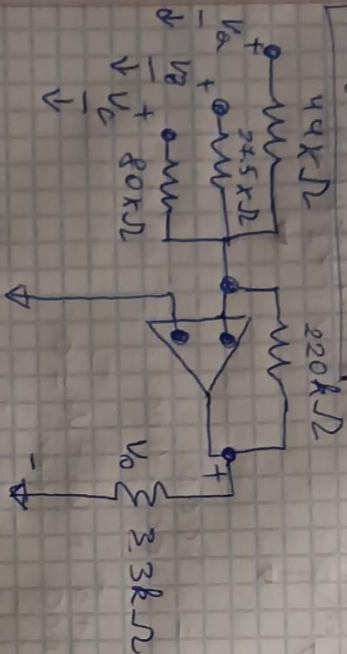
p. 165 ex. 5.4



$$\frac{V_b - V_a}{5000} + \frac{V_b - V_o}{40000} = 0$$

- a) $V_a = 1,5V$ $V_b = 0V$ $V_o = 12V$
 b) $V_a = -0,5V$ $V_b = 0V$ $V_o = 4V$
 c) $V_a = 1V$ $V_b = 2,5V$ $V_o = 14,5V$
 d) $V_a = 2,5V$ $V_b = 1V$ $V_o = -14V$
 e) $V_a = 2,5V$ $V_b = 0V$ $V_o = -20V$

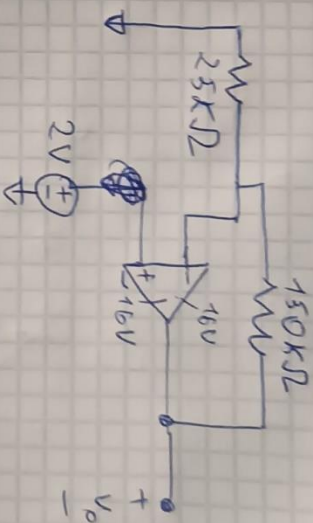
P. 166 ex. 5.12



a) Example of an inverting summing amplifier

b) $V_o = \frac{220}{44} V_a - \frac{220}{24.5} V_b - \frac{220}{3.3} V_c = -5 - 12 + 4 = -6V$

P. 164 ex. 5.18



a) non-inverting amplifier

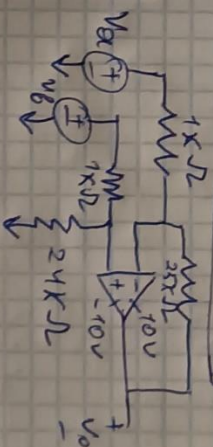
ideal $V_n = V_p = 2V$

KCL at V_n : $\frac{2}{25000} + \frac{2 - V_o}{150000} = 0$

$12 + 2 - V_o = 0$

$V_o = 14V$

P. 168 ex. 5.33



$A_{dm} = \frac{24 \cdot 26 + 28 \cdot 25}{2 \cdot 1 \cdot 25} = 24,88$